

```
!pip install transformers
!pip install torch
```

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Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.12.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.3.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.24.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (23.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.1)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2023.10.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.12.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.4.3)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.66.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (2023.12.2)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.4.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.23.1)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub) (0.12.1)
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (2023.11.17)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (1.0.2)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*) (0.14.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer) (8.1.7)
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer) (13.3.1)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer) (0.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich) (2.17.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py) (0.1.1)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.12.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.5.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (68.0.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.12)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.2)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2023.12.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2) (2.1.2)
```

```
from transformers import pipeline
import torch

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM, pipeline

# Directly load the tokenizer and model, bypassing the pipeline's task registry issue
tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn")

# Create a custom callable summarizer object that mimics the pipeline's behavior
class CustomSummarizer:
    def __init__(self, model, tokenizer):
        self.model = model
        self.tokenizer = tokenizer

    def __call__(self, text, max_length=150, min_length=30, do_sample=False):
        inputs = self.tokenizer(
            text,
            max_length=1024,
            return_tensors="pt",
            truncation=True
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    )
    summary_ids = self.model.generate(
        inputs["input_ids"],
        num_beams=4,
        max_length=max_length,
        min_length=min_length,
        do_sample=do_sample,
        early_stopping=True
    )
    # The pipeline typically returns a list of dictionaries, e.g., [{'summary_text': '...'}]
    return [{"summary_text": self.tokenizer.decode(g, skip_special_tokens=True, clean_up_token_spaces=True)} for g in summary_ids]

summarizer = CustomSummarizer(model, tokenizer)

print("Model loaded successfully!")

```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable faster downloads.
 WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub, which may result in slower download speeds.
 vocab.json: 899k/? [00:00<00:00, 13.5MB/s]
 merges.txt: 456k/? [00:00<00:00, 6.61MB/s]
 tokenizer.json: 1.36M/? [00:00<00:00, 17.6MB/s]
 model.safetensors: 100% 1.63G/1.63G [00:21<00:00, 73.3MB/s]
 Please make sure the generation config includes `forced\_bos\_token\_id=0`.
 Loading weights: 100% 511/511 [00:01<00:00, 573.71it/s, Materializing param=model.encoder.layers.11.self_attn_layer_norm]
 generation_config.json: 100% 363/363 [00:00<00:00, 18.6kB/s]
 Model loaded successfully!

```

texts = [
    """Artificial Intelligence (AI) is transforming industries by enabling machines
    to learn from data and make decisions. It is used in healthcare, finance,
    autonomous vehicles, and more. AI improves efficiency but also raises ethical concerns
    such as job displacement and data privacy.""",

    """The Indian Space Research Organisation (ISRO) has successfully launched multiple satellites.
    Its cost-effective missions have gained global recognition. Chandrayaan and Mangalyaan
    are major achievements that demonstrate India's growing capabilities in space technology.""",

    """Once upon a time, there was a young boy who dreamed of exploring the world.
    He faced many challenges but never gave up. His journey taught him valuable lessons
    about courage, persistence, and kindness.""",

    """Machine learning models require large amounts of data. Training involves optimizing
    parameters to minimize error. Deep learning models, especially neural networks,
    are computationally expensive but powerful.""",

    """Hey, did you finish the assignment? Not yet, I was planning to complete it tonight.
    The deadline is tomorrow, so we should hurry. Let's work together and finish it quickly.""
]

print("Dataset loaded with", len(texts), "texts")

```

Dataset loaded with 5 texts

```

for i, text in enumerate(texts):
    summary = summarizer(text, max_length=60, min_length=20, do_sample=False)

    print(f"\n--- TEXT {i+1} ---")
    print("Original:", text)
    print("Summary:", summary[0]['summary_text'])

```

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--- TEXT 1 ---
Original: Artificial Intelligence (AI) is transforming industries by enabling machines
to learn from data and make decisions. It is used in healthcare, finance,
autonomous vehicles, and more. AI improves efficiency but also raises ethical concerns
such as job displacement and data privacy.
Summary: Artificial Intelligence (AI) is transforming industries by enabling machines to learn from

--- TEXT 2 ---
Original: The Indian Space Research Organisation (ISRO) has successfully launched multiple satellites.
Its cost-effective missions have gained global recognition. Chandrayaan and Mangalyaan
are major achievements that demonstrate India's growing capabilities in space technology.
Summary: The Indian Space Research Organisation (ISRO) has successfully launched multiple satellites

--- TEXT 3 ---
Original: Once upon a time, there was a young boy who dreamed of exploring the world.
He faced many challenges but never gave up. His journey taught him valuable lessons
about courage, persistence, and kindness.
Summary: Once upon a time, there was a young boy who dreamed of exploring the world. He faced many

--- TEXT 4 ---
Original: Machine learning models require large amounts of data. Training involves optimizing
parameters to minimize error. Deep learning models, especially neural networks,
are computationally expensive but powerful.
Summary: Machine learning models require large amounts of data. Training involves optimizing parameters

--- TEXT 5 ---
Original: Hey, did you finish the assignment? Not yet, I was planning to complete it tonight.
The deadline is tomorrow, so we should hurry. Let's work together and finish it quickly.
Summary: Hey, did you finish the assignment? Not yet, I was planning to complete it tonight. The c

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```

text = texts[0]

short_summary = summarizer(text, max_length=30, min_length=10, do_sample=False)
medium_summary = summarizer(text, max_length=60, min_length=20, do_sample=False)
long_summary = summarizer(text, max_length=100, min_length=40, do_sample=False)

print("\nShort Summary:\n", short_summary[0]['summary_text'])
print("\nMedium Summary:\n", medium_summary[0]['summary_text'])
print("\nLong Summary:\n", long_summary[0]['summary_text'])

```

```

Short Summary:
Artificial Intelligence (AI) is transforming industries by enabling machines to learn from data.

Medium Summary:
Artificial Intelligence (AI) is transforming industries by enabling machines to learn from data.

Long Summary:
Artificial Intelligence (AI) is transforming industries by enabling machines to learn from data.

```

```

long_text = """Artificial Intelligence has rapidly evolved over the past decade.
It is now used in various sectors including healthcare, education, transportation,
and finance. AI systems can analyze massive datasets and provide insights that
were previously impossible. However, there are challenges such as bias in data,
lack of transparency, and ethical concerns. Governments and organizations are
working to regulate AI and ensure responsible usage. Future advancements may
lead to even more powerful systems that can outperform humans in complex tasks."""

summary = summarizer(long_text, max_length=80, min_length=30, do_sample=False)

print("Long Text Summary:\n", summary[0]['summary_text'])

```

```

Long Text Summary:
Artificial Intelligence has rapidly evolved over the past decade. It is now used in various sectors

```

